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Interactive Visualization of Research Projects with R-Shiny based on a by-catch study

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Deutsches Meeresmuseum

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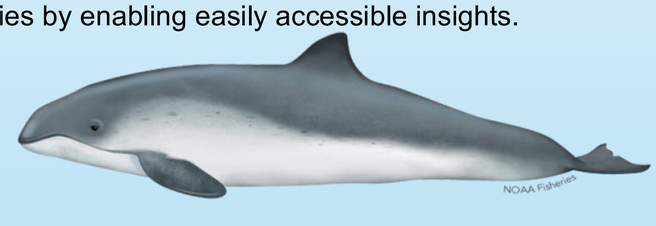
Abstract

Presenting research results and project setups can be challenging, particularly when the project pipeline involves multiple stages, such as data preparation, exploration, modeling, or other specific analyses. However, presenting data in an understandable and easily accessible manner fosters transparency and outreach. To address this, an interactive dashboard, built using the shiny and golem packages in R, has been developed to enhance the visualization and communication of research findings in the PAL-CE project by the German Oceanographic Museum. This R-based dashboard not only showcases the effectiveness of the PAL but also demonstrates how interactive tools can streamline the presentation of complex ecological studies by enabling easily accessible insights.

Case study: PAL-CE project

The project investigates the impact of Porpoise ALerts (PAL) on harbour porpoises in the Baltic Sea. PAL was designed to mitigate bycatch by emitting artificial, porpoise-like sounds to increase the animals' echolocation activity. Data from four field campaigns over three years in Denmark and Germany, was collected using several **FPODs** and **Soundtraps** for up to three months per campaign. These devices monitored porpoises under different scenarios. The acoustic and environmental datasets were analysed with generalized additive models (GAMs) to evaluate the impact of different variables. Current results suggest that PAL significantly increases porpoise click rates near fishing nets, effectively alerting the animals without displacing them from the area.



About Shiny and Golem

Shiny is an R package that makes it easy to build interactive web applications straight from R. Shiny apps are contained in a unique script call app.R, which has 2 components: **UI** and **Server**.

Golem provides a framework to create and maintain **production-grade** Shiny apps. Useful when the app becomes bigger and more complex.

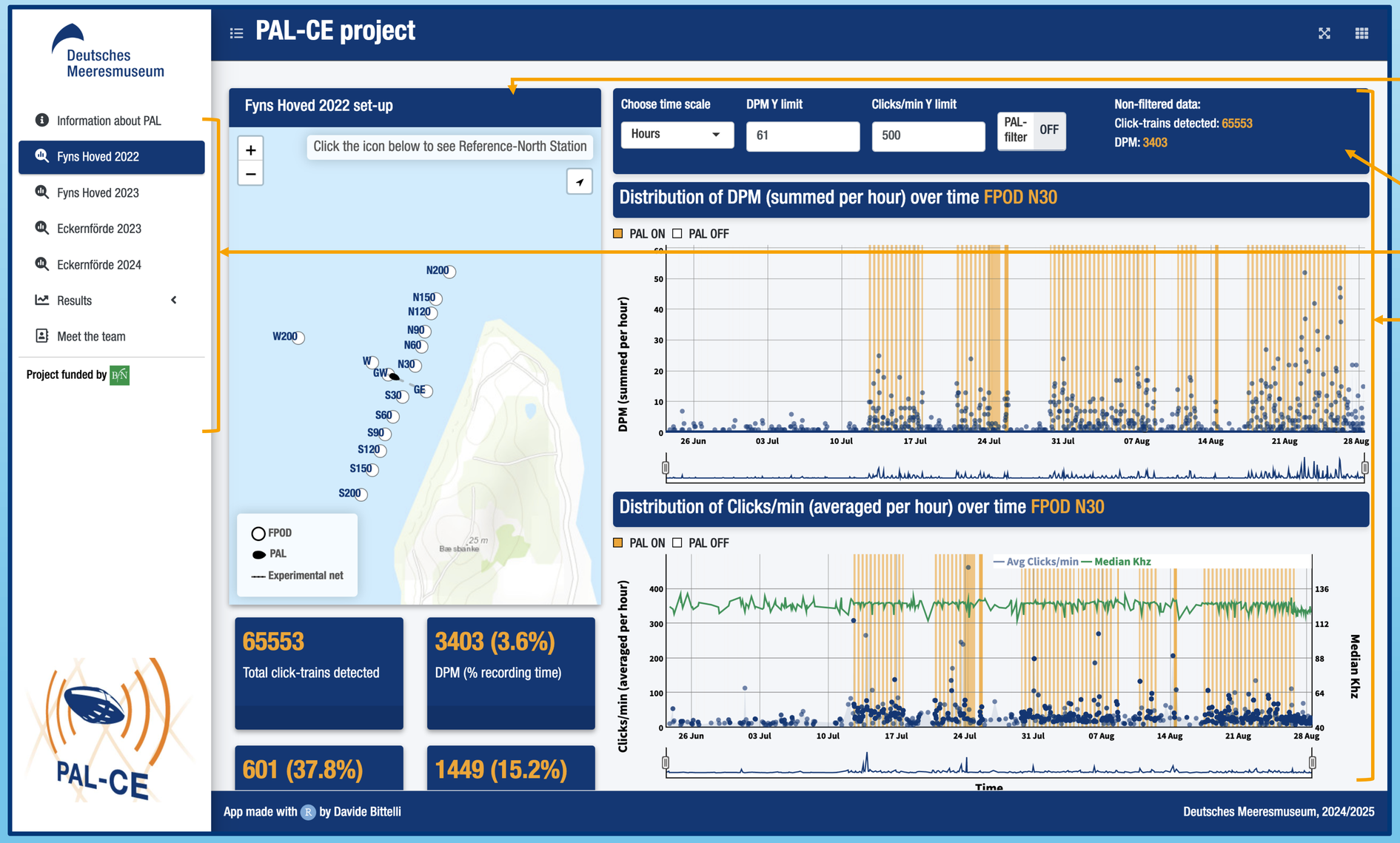
Modules and ns()

Modules are pieces of a Shiny app. They are self-contained building blocks encapsuling specific features of the app. Each module has its UI function and server function.

Namespaces make sure in/output IDs within the modules don't collide.

App Structure

```
|── DESCRIPTION
|── NAMESPACE
|── R
|    |── app_config.R
|    |── app_server.R
|    |── app_ui.R
|    └── run_app.R
|── dev
|    |── 01_start.R
|    |── 02_dev.R
|    |── 03_deploy.R
|    └── run_dev.R
|── inst
|    |── app
|    |     └── www
|    └── golem-congi.yml
└── man
      └── run_app.Rd
```



Customization

Map is a Module, F-PODs have ns()
Interactivity across app
leaflet() + **CSS** + **JS**

Control Panel
Shiny + **HTML** + **CSS**

Modules
Shiny

HTML object containing Interactive headers and plots. Fed by a fct()
Depending on map ns()
R plots + **HTML** + **CSS**

Useful functions()

golem::
add_module()
add_fct()
add_css_file()
add_js_file()

attachment::
att_amend_desc()

Durability and Reproducibility

The **renv** package ensures that an R Shiny app always uses the exact same package versions, preventing issues caused by package updates or system differences. This makes the app stable, reliable, and easily reproducible over time.

While developing

```
renv::init()        # initialize new project
renv::snapshot()    # save state of the libraries & dependencies
```

When restoring

```
renv::restore()     # restore project libraries & dependencies
```

Deployment on the Web

Deploying an R Shiny app makes it accessible online to users or collaborators.

Cloud
Easy but with limitations

- **shinyapps.io**
- **Posit Connect**
- **Shiny Server**

Self Managed
Complex but more flexible

- **Docker**
- **Shinylive + webR + GitHub pages**
- **R Package**

Open the App on your **Phone**

Deployed w **shinyapps.io**

Acknowledgements

- **PAL-CE** Team
- **ThinkR**
- Book ***Engineering Production-Grade Shiny Apps*** (Colin Fay, Sébastien Rochette, Vincent Guyader and Cervan Girard)